

Supplementary Information for MedEquiSeg:
Shared Augmentation and Privileged-Prompt
Reliability in Multimodal Medical Image
Segmentation

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Layer-resolved MedEquiSeg architecture

The main manuscript shows the effective forward graph at a readable scale. The following two figures expose the source-verified layer transitions and custom operators. They do not add dormant modules or token-level fusion to the evaluated graph.

Boundary and empty-mask metrics

All entries were recomputed after predictions were restored to the native reference-mask grid with the common evaluation procedure. A both-empty case receives perfect overlap and zero distance. A one-sided empty case receives zero overlap and infinite distance. For interpretability, the tables report finite-case HD95 and ASSD together with the one-empty failure rate; finite distance values must not be interpreted without the corresponding failure rate. NSD uses a two-pixel tolerance.

Table S1 Boundary and empty-mask metrics for BUSI under the common evaluation procedure. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	41.11 $\pm$ 0.66	32.64 $\pm$ 6.09	10.92 $\pm$ 1.67	0.00 $\pm$ 0.00
CausalCLIPSeg	43.58 $\pm$ 0.50	40.61 $\pm$ 3.01	12.55 $\pm$ 0.97	0.00 $\pm$ 0.00
LViT	41.26 $\pm$ 1.45	48.09 $\pm$ 2.87	14.39 $\pm$ 0.67	1.28 $\pm$ 1.28
MEDEquiSEG (Ours)	43.96 $\pm$ 1.30	35.86 $\pm$ 3.44	11.42 $\pm$ 0.92	0.00 $\pm$ 0.00
RollingUNet [†]	29.81 $\pm$ 2.29	65.72 $\pm$ 2.14	20.51 $\pm$ 1.38	7.26 $\pm$ 1.96
U-Net [†]	12.27 $\pm$ 1.86	120.79 $\pm$ 31.04	30.80 $\pm$ 6.69	16.67 $\pm$ 0.00
U-Net++ [†]	34.69 $\pm$ 1.46	48.76 $\pm$ 3.25	15.78 $\pm$ 0.46	6.41 $\pm$ 1.28
nnU-Net [†]	33.76 $\pm$ 0.52	44.65 $\pm$ 2.83	15.43 $\pm$ 2.08	8.55 $\pm$ 1.96
U-Mamba [†]	35.48 $\pm$ 1.47	49.16 $\pm$ 12.17	16.64 $\pm$ 2.79	6.84 $\pm$ 0.74
U-KAN [†]	37.10 $\pm$ 2.93	48.21 $\pm$ 2.73	16.79 $\pm$ 1.73	5.98 $\pm$ 2.96

COVID-19 subject-cluster sensitivity

This post hoc sensitivity analysis holds the reported model parameters fixed. It accounts for repeated test images from 474 recoverable subjects, but it does not repair

Table S2 Boundary and empty-mask metrics for ClinicDB under the common evaluation procedure. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	58.50 $\pm$ 1.19	12.25 $\pm$ 1.07	3.40 $\pm$ 0.15	0.00 $\pm$ 0.00
CausalCLIPSeg	64.26 $\pm$ 2.37	15.78 $\pm$ 1.04	3.84 $\pm$ 0.28	0.00 $\pm$ 0.00
LViT	65.26 $\pm$ 1.48	16.28 $\pm$ 3.10	3.96 $\pm$ 0.71	0.00 $\pm$ 0.00
MEDEQUISEG (Ours)	67.00 $\pm$ 0.24	14.99 $\pm$ 1.08	3.61 $\pm$ 0.34	0.00 $\pm$ 0.00
RollingUNet [†]	49.88 $\pm$ 2.32	21.46 $\pm$ 1.76	5.19 $\pm$ 0.18	1.09 $\pm$ 0.95
U-Net [†]	38.83 $\pm$ 3.45	36.13 $\pm$ 5.40	8.85 $\pm$ 1.41	0.00 $\pm$ 0.00
U-Net++ [†]	60.07 $\pm$ 1.13	18.47 $\pm$ 1.58	4.75 $\pm$ 0.20	0.00 $\pm$ 0.00
nnU-Net [†]	70.58 $\pm$ 0.48	9.82 $\pm$ 1.23	2.71 $\pm$ 0.25	0.00 $\pm$ 0.00
U-Mamba [†]	69.79 $\pm$ 1.25	13.23 $\pm$ 0.52	3.60 $\pm$ 0.22	0.00 $\pm$ 0.00
U-KAN [†]	62.93 $\pm$ 0.33	15.17 $\pm$ 0.94	3.89 $\pm$ 0.18	0.00 $\pm$ 0.00

Table S3 Boundary and empty-mask metrics for BUS-BRA under the common evaluation procedure. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	45.54 $\pm$ 0.76	16.13 $\pm$ 1.54	5.09 $\pm$ 0.66	0.00 $\pm$ 0.00
CausalCLIPSeg	49.57 $\pm$ 1.27	15.53 $\pm$ 0.67	4.99 $\pm$ 0.42	0.00 $\pm$ 0.00
LViT	50.40 $\pm$ 0.61	20.00 $\pm$ 0.82	5.84 $\pm$ 0.30	0.00 $\pm$ 0.00
MEDEQUISEG (Ours)	52.53 $\pm$ 0.29	13.81 $\pm$ 0.59	4.31 $\pm$ 0.15	0.00 $\pm$ 0.00
RollingUNet [†]	50.67 $\pm$ 1.21	18.71 $\pm$ 1.34	5.51 $\pm$ 0.35	0.12 $\pm$ 0.20
U-Net [†]	44.75 $\pm$ 0.66	22.17 $\pm$ 0.69	6.05 $\pm$ 0.11	0.00 $\pm$ 0.00
U-Net++ [†]	53.04 $\pm$ 0.20	18.50 $\pm$ 0.52	5.22 $\pm$ 0.05	0.00 $\pm$ 0.00
nnU-Net [†]	53.89 $\pm$ 0.41	14.83 $\pm$ 0.61	4.52 $\pm$ 0.19	0.00 $\pm$ 0.00
U-Mamba [†]	55.25 $\pm$ 0.21	13.72 $\pm$ 0.74	4.32 $\pm$ 0.26	0.00 $\pm$ 0.00
U-KAN [†]	52.63 $\pm$ 0.68	16.98 $\pm$ 0.32	5.19 $\pm$ 0.29	0.00 $\pm$ 0.00

the known overlap of recoverable subject IDs between the fixed training and validation folders.

Realized rewrite-trigger frequencies

The following table applies the deterministic MEDEQUISEG augmentation procedure to every training case, epoch, and seed without loading image content. It reports realized rather than analytically expected trigger frequencies.

Table S4 Boundary and empty-mask metrics for BRISC under the common evaluation procedure. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	48.90 $\pm$ 4.84	15.25 $\pm$ 0.91	5.82 $\pm$ 0.70	0.90 $\pm$ 0.40
CausalCLIPSeg	65.90 $\pm$ 1.43	14.69 $\pm$ 1.46	4.82 $\pm$ 0.32	0.49 $\pm$ 0.75
LViT	70.78 $\pm$ 0.45	14.82 $\pm$ 0.50	4.37 $\pm$ 0.06	1.64 $\pm$ 0.34
MEDEQUISEG (Ours)	67.12 $\pm$ 0.16	12.63 $\pm$ 0.20	4.23 $\pm$ 0.30	0.37 $\pm$ 0.17
RollingUNet [†]	64.64 $\pm$ 0.72	13.60 $\pm$ 0.24	4.67 $\pm$ 0.25	1.64 $\pm$ 0.36
U-Net [†]	61.68 $\pm$ 0.98	13.82 $\pm$ 0.31	4.87 $\pm$ 0.11	0.64 $\pm$ 0.23
U-Net++ [†]	67.66 $\pm$ 0.47	13.55 $\pm$ 0.16	4.75 $\pm$ 0.17	0.75 $\pm$ 0.32
nnU-Net [†]	73.33 $\pm$ 0.51	12.58 $\pm$ 0.54	3.95 $\pm$ 0.26	1.49 $\pm$ 0.39
U-Mamba [†]	70.19 $\pm$ 1.10	13.23 $\pm$ 0.30	4.62 $\pm$ 0.18	2.02 $\pm$ 0.30
U-KAN [†]	64.89 $\pm$ 0.53	13.02 $\pm$ 0.69	4.28 $\pm$ 0.18	1.57 $\pm$ 0.40

Table S5 Boundary and empty-mask metrics for COVID-19 under the common evaluation procedure. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	67.37 $\pm$ 2.09	18.10 $\pm$ 3.18	4.45 $\pm$ 0.65	0.00 $\pm$ 0.00
CausalCLIPSeg	68.24 $\pm$ 0.19	20.02 $\pm$ 0.13	5.50 $\pm$ 0.09	0.08 $\pm$ 0.07
LViT	65.75 $\pm$ 0.54	27.13 $\pm$ 0.55	6.96 $\pm$ 0.15	0.65 $\pm$ 0.36
MEDEQUISEG (Ours)	71.08 $\pm$ 0.26	15.03 $\pm$ 0.71	3.72 $\pm$ 0.18	0.06 $\pm$ 0.03
RollingUNet [†]	66.47 $\pm$ 0.36	29.17 $\pm$ 0.76	6.86 $\pm$ 0.09	0.19 $\pm$ 0.09
U-Net [†]	64.08 $\pm$ 0.45	30.47 $\pm$ 0.20	7.13 $\pm$ 0.08	0.00 $\pm$ 0.00
U-Net++ [†]	66.57 $\pm$ 0.15	28.02 $\pm$ 0.24	6.53 $\pm$ 0.05	0.00 $\pm$ 0.00
nnU-Net [†]	66.09 $\pm$ 0.31	28.01 $\pm$ 0.65	6.55 $\pm$ 0.16	0.71 $\pm$ 0.14
U-Mamba [†]	66.35 $\pm$ 0.43	27.78 $\pm$ 0.73	6.70 $\pm$ 0.18	0.62 $\pm$ 0.16
U-KAN [†]	67.23 $\pm$ 0.31	29.32 $\pm$ 1.00	6.88 $\pm$ 0.23	0.38 $\pm$ 0.17

Table S6 COVID-19 subject-cluster sensitivity analysis on the fixed test set. The image-weighted delta is the main evaluator point estimate, and its 95% interval resamples recoverable subjects and training seeds with 10,000 replicates. Subject-weighted deltas first average the three-seed difference within each of 474 subjects and then weight subjects equally. p_{subj} is the two-sided Wilcoxon test on those equal-subject mean differences. Thus, the confidence interval and p -value describe explicitly different estimands. All effects are percentage points.

Comparison	Metric	Image Δ	Cluster 95% CI	Subject Δ	p_{subj}
MEDEQUISEG – U-Net++	Dice	+4.08	[3.48, 4.72]	+5.30	< .001
	IoU	+5.00	[4.30, 5.68]	+5.95	< .001
True – shuffled	Dice	+15.29	[13.99, 16.66]	+17.68	< .001
	IoU	+18.04	[16.61, 19.51]	+19.39	< .001

Table S7 Realized MEDEQUISEG rewrite-trigger frequencies over all training cases, 100 epochs, and seeds 123, 456, and 789. Spatial coverage is the percentage of training prompts containing a recognized direction phrase. Discrete-transformation rate counts samples receiving at least one flip or right-angle rotation; string-change rate counts samples whose final prompt differs from its input after the complete composed rewrite. Values for event rates are mean $\pm$ sample standard deviation across seeds.

Dataset	Train	Events/seed	Spatial (%)	Discrete transform. (%)	String change (%)
BUSI	624	62400	82.85	48.42 ± 0.03	20.27 ± 0.10
ClinicDB	490	49000	100.00	48.37 ± 0.24	28.48 ± 0.21
BUS-BRA	1311	131100	100.00	48.42 ± 0.20	12.44 ± 0.12
BRISC	3881	388100	78.23	48.49 ± 0.12	5.75 ± 0.04
COVID-19	5716	571600	100.00	48.54 ± 0.04	45.75 ± 0.06

Mask-presence sensitivity

This analysis separates lesion-containing cases from empty-reference-mask cases so that correct empty predictions do not implicitly stand in for lesion-boundary quality.

Full ordered configuration results

The main manuscript reports the compact five-dataset macro-average. The full dataset-level matrix is retained here to show that the macro difference is not uniform across datasets and to preserve all 75 retained dataset–configuration–seed combinations. The sequence is not a full crossing of adapters, augmentation policy, and rewriting; row differences are therefore conditional configuration contrasts rather than independent component effects.

COVID-19 grouped train–validation sensitivity

The fixed public test set is unchanged in this analysis. Recoverable subject tokens are assigned to only one training or validation split; rows without a recoverable token remain case-level groups. This is a split-sensitivity analysis, not evidence that every image in the source release has a known patient identity.

Table S8 Mask-presence sensitivity analysis under the common evaluation procedure. Non-empty Dice and IoU are computed only on test images containing foreground. Empty specificity is the percentage of empty-reference-mask images predicted as empty. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. A dash indicates that the test split contains no empty masks. The image-only reference is selected by the highest overall three-seed Dice within each dataset before this subgroup analysis.

Dataset	Method	Dice	IoU	Empty specificity
BUSI	MEDEQUISEG (Ours)	82.94 $\pm$ 0.55	75.39 $\pm$ 0.28	100.00 $\pm$ 0.00
	MedCLIPSeg	83.36 $\pm$ 1.30	75.22 $\pm$ 1.34	100.00 $\pm$ 0.00
	U-KAN [†]	79.70 $\pm$ 0.48	71.63 $\pm$ 0.57	64.10 $\pm$ 17.76
ClinicDB	MEDEQUISEG (Ours)	92.07 $\pm$ 0.71	86.36 $\pm$ 0.61	–
	MedCLIPSeg	91.96 $\pm$ 0.07	85.64 $\pm$ 0.10	–
	nnU-Net [†]	93.62 $\pm$ 0.31	88.63 $\pm$ 0.29	–
BUS-BRA	MEDEQUISEG (Ours)	90.24 $\pm$ 0.14	83.15 $\pm$ 0.10	–
	MedCLIPSeg	88.47 $\pm$ 0.66	80.46 $\pm$ 0.76	–
	U-Mamba [†]	90.34 $\pm$ 0.46	83.58 $\pm$ 0.47	–
BRISC	MEDEQUISEG (Ours)	84.82 $\pm$ 0.17	76.68 $\pm$ 0.16	100.00 $\pm$ 0.00
	MedCLIPSeg	77.92 $\pm$ 2.12	67.36 $\pm$ 2.67	100.00 $\pm$ 0.00
	nnU-Net [†]	85.08 $\pm$ 0.34	77.99 $\pm$ 0.43	99.28 $\pm$ 0.72
COVID-19	MEDEQUISEG (Ours)	84.08 $\pm$ 0.13	75.51 $\pm$ 0.14	–
	MedCLIPSeg	82.62 $\pm$ 1.19	73.47 $\pm$ 1.48	–
	U-Net++ [†]	80.00 $\pm$ 0.06	70.51 $\pm$ 0.08	–

[†]Image-only contextual reference; cross-modality differences are descriptive because public benchmark prompts contain target-derived attributes.

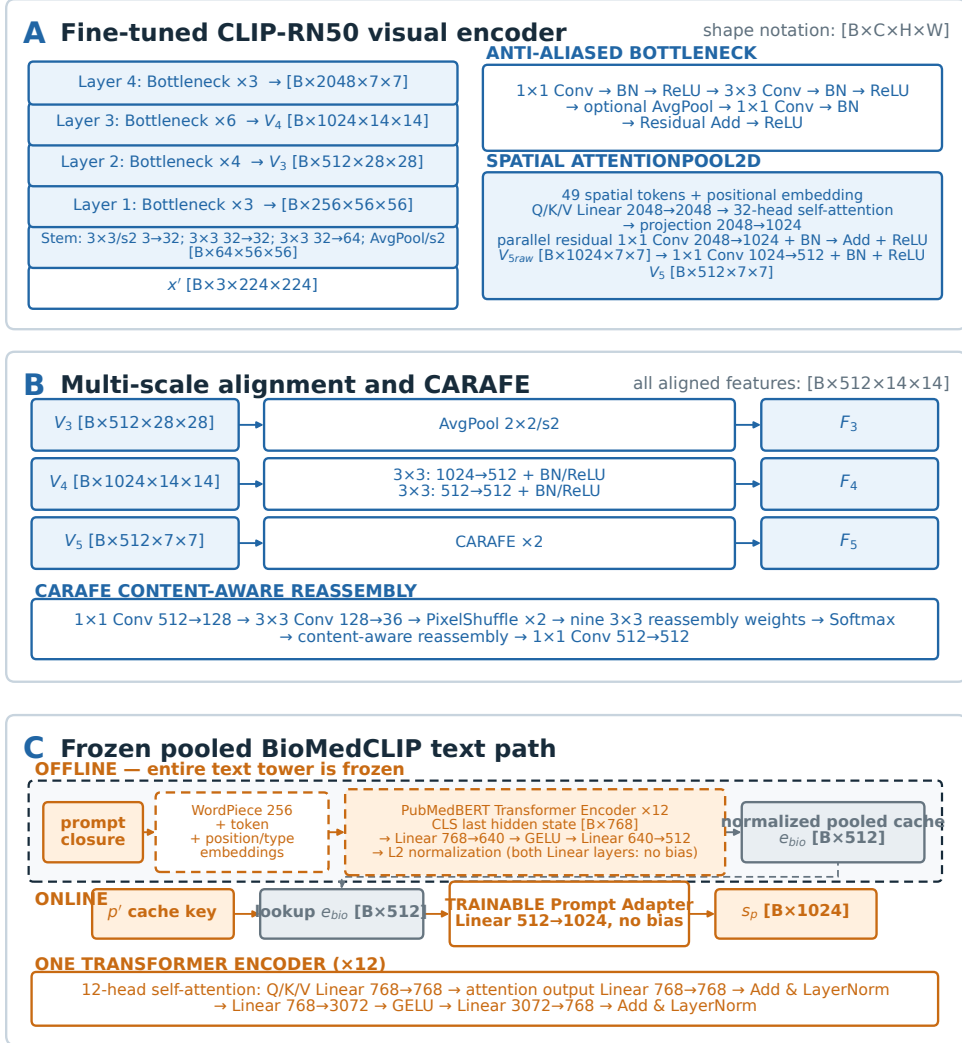


Fig. S1 Layer-resolved encoder and text representation. (A) The fine-tuned OpenAI CLIP-RN50 stage stack, anti-aliased bottleneck, and spatial-only AttentionPool2d implementation. (B) The three multi-scale alignment routes and the CARAFE reassembly operator. (C) The complete frozen BioMedCLIP text tower, the boundary between one-time cache construction and online lookup, and the trainable pooled-feature adapter. BioMedCLIP contributes a pooled 512-D vector; token-level text fusion is inactive.

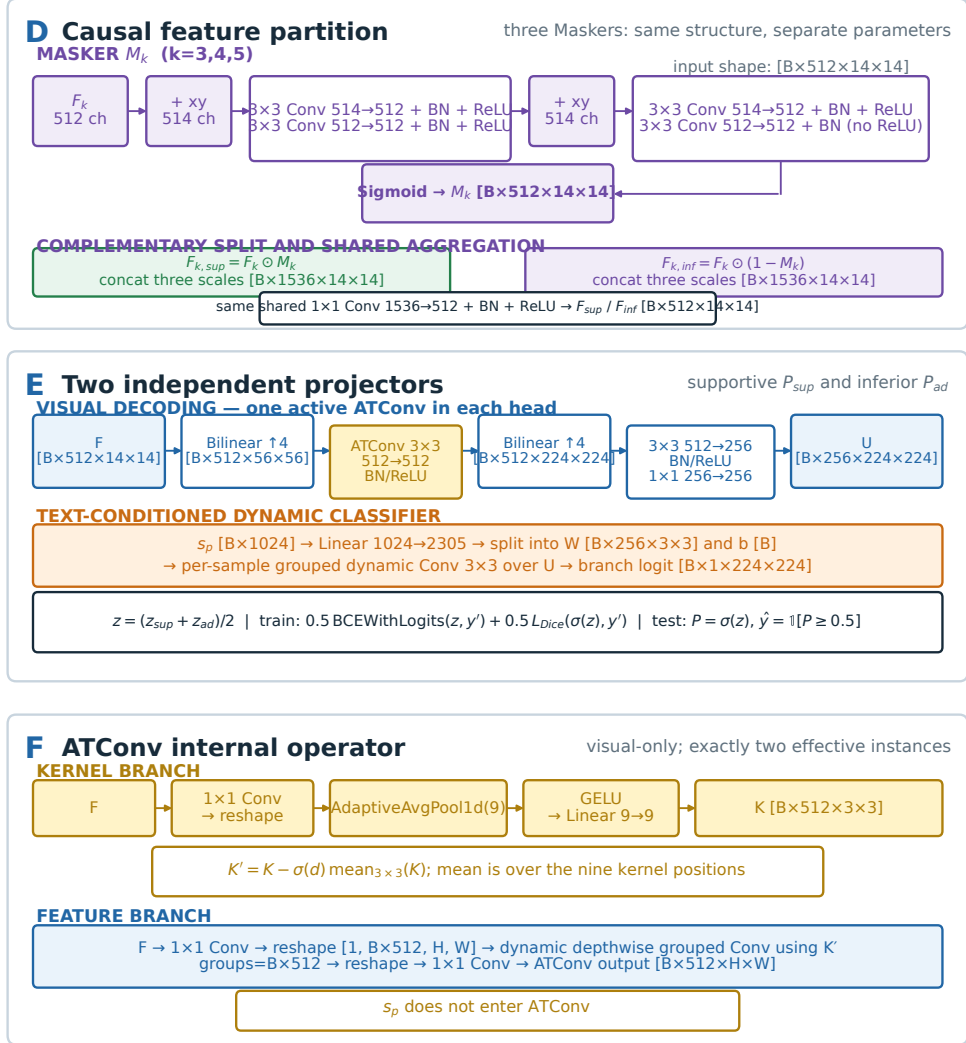


Fig. S2 Layer-resolved custom operators. (D) Each scale has a structurally identical but independently parameterized Masker; supportive and inferior features are complementary, and their three-scale aggregation uses shared weights. (E) The two projectors are parameter-independent and each contains one active ATConv followed by a text-conditioned per-sample dynamic classifier. (F) ATConv’s visual kernel and feature branches. The pooled semantic state does not enter ATConv itself.

Table S9 Per-dataset results for five retained ordered configurations. Each configuration was evaluated with three random seeds under common data partitions and evaluation procedures. ATConv denotes the two projection-head operators active in the evaluated forward pass; shared augmentation and prompt rewriting appear together only in the final configuration. Values are Dice and IoU percentages (mean $\pm$ sample standard deviation). Because the joint final addition is not crossed with every adapter combination, row differences are conditional configuration contrasts, not independent component effects. Bold and underline identify the best and second-best configuration within each dataset and metric.

Configuration	Metric	BUSI	ClinicDB	BUS-BRA
Base	Dice	84.50 $\pm$ 0.82	90.15 $\pm$ 0.97	89.26 $\pm$ 0.14
	IoU	77.86 $\pm$ 0.80	84.19 $\pm$ 0.91	<u>81.83 $\pm$ 0.22</u>
+ BioMedCLIP	Dice	83.76 $\pm$ 0.47	90.47 $\pm$ 1.03	89.17 $\pm$ 0.17
	IoU	76.96 $\pm$ 0.59	84.88 $\pm$ 0.96	81.67 $\pm$ 0.22
+ ATConv	Dice	84.76 $\pm$ 0.48	90.87 $\pm$ 0.99	89.19 $\pm$ 0.42
	IoU	78.08 $\pm$ 0.26	84.88 $\pm$ 1.17	81.78 $\pm$ 0.53
+ BioMedCLIP + ATConv	Dice	84.87 $\pm$ 0.46	91.49 $\pm$ 0.47	89.08 $\pm$ 0.22
	IoU	<u>78.08 $\pm$ 0.51</u>	<u>85.57 $\pm$ 0.56</u>	81.76 $\pm$ 0.23
MedEquiSeg (Ours)	Dice	85.78 $\pm$ 0.45	92.07 $\pm$ 0.71	90.24 $\pm$ 0.14
	IoU	79.49 $\pm$ 0.23	86.36 $\pm$ 0.61	83.15 $\pm$ 0.10
Configuration	Metric	BRISC	COVID-19	Macro avg.
Base	Dice	86.52 $\pm$ 0.17	81.65 $\pm$ 0.08	86.41 $\pm$ 0.35
	IoU	79.52 $\pm$ 0.14	72.71 $\pm$ 0.14	79.22 $\pm$ 0.34
+ BioMedCLIP	Dice	<u>86.58 $\pm$ 0.06</u>	83.35 $\pm$ 0.07	86.67 $\pm$ 0.08
	IoU	<u>79.60 $\pm$ 0.04</u>	<u>74.49 $\pm$ 0.09</u>	79.52 $\pm$ 0.06
+ ATConv	Dice	86.34 $\pm$ 0.11	82.02 $\pm$ 0.16	86.63 $\pm$ 0.32
	IoU	79.31 $\pm$ 0.14	73.06 $\pm$ 0.11	79.42 $\pm$ 0.31
+ BioMedCLIP + ATConv	Dice	86.39 $\pm$ 0.25	83.32 $\pm$ 0.10	87.03 $\pm$ 0.15
	IoU	79.41 $\pm$ 0.18	74.42 $\pm$ 0.19	<u>79.85 $\pm$ 0.23</u>
MedEquiSeg (Ours)	Dice	87.18 $\pm$ 0.14	84.08 $\pm$ 0.13	87.87 $\pm$ 0.17
	IoU	80.32 $\pm$ 0.13	75.51 $\pm$ 0.14	80.97 $\pm$ 0.13

Table S10 COVID-19 grouped train-validation sensitivity. The fixed test set is unchanged. The sensitivity partition contains 5,706 training, 1,439 validation, and 2,113 test images; overlap among 2,130 recoverable subject tokens is zero across all splits. Values are percentages (mean $\pm$ sample standard deviation over seeds 123, 456, and 789). Δ is the grouped-validation result minus a separately trained, configuration-matched original-split sensitivity result in percentage points; these sensitivity models are distinct from those used in the main ordered configuration analysis.

Method	Dice	Original	Δ	IoU	Original	Δ
MEDEQUISEG	83.99 $\pm$ 0.07	84.18 $\pm$ 0.13	-0.19	75.35 $\pm$ 0.09	75.64 $\pm$ 0.14	-0.28
U-Net++	80.14 $\pm$ 0.09	80.00 $\pm$ 0.05	+0.14	70.68 $\pm$ 0.18	70.51 $\pm$ 0.10	+0.17

Reproducibility and implementation verification

This verification distinguishes the fixed data and evaluation procedures from the later shared-augmentation experiment and examines the effective ATConv execution graph. It is descriptive of the reported experiments and does not alter model parameters or predictions.

Table S11 Reproducibility and implementation checks for the reported experiments.

Verification item	Evidence	Interpretation
Data and evaluation consistency	Dataset partitions, split roles, mask processing, random seeds, prediction threshold, and evaluation metrics were constant across comparisons; the training policy is documented separately.	Supports comparisons under common data and evaluation procedures, but not a claim that all training choices were fixed prospectively.
Shared augmentation policy	The initial policy restricted ClinicDB and COVID-19 to horizontal flipping and excluded continuous geometric transformations. The shared policy applies multiple geometric transformations plus mild affine, translation, and zoom operations to all datasets.	Introduced after the initial augmentation specification; treated as a retrospective follow-up, not a prospectively specified factorial component.
Active ATConv path	The model contains four registered replacements. Module-activation monitoring on a real BUSI test case recorded one non-zero output for each projector replacement and no activation of the two neck modules.	The effective forward pass contains two active projector ATConv operators.
Experiment traceability	Records link the data partitions, source files, text cache, training settings, epoch, and selected model. The summary omits batch size and worker count, but supplementary execution records retain them. Identity checks for all 75 retained models agreed with the archived records.	Results remain model-traceable; the omitted settings are disclosed as a reproducibility limitation.
Source reconstruction	The archived source package omits four orchestration and evaluation files, and the available release copies cannot be verified as identical to the source used for the aggregate experiments.	Exact reconstruction of the original software environment is not guaranteed.
Training-log completeness	Seventy-four of 75 retained logs contain 100 epoch records with consistent model-selection records. One intermediate BioMedCLIP-ATConv COVID-19 seed-789 log was overwritten during an unintended repeated execution; its metadata and four subsequent verification analyses identify the same selected model.	The numerical result remains model-traceable, but the associated training log is incomplete.

Exploratory continuous-location diagnostic

This exploratory experiment uses the same transformation distribution but an independently initialized augmentation sequence. It is therefore retained only as a distribution-matched analysis, not a paired mechanism comparison.

Table S12 Exploratory three-seed continuous-location recomputation analysis. MEDEQUISEG + Location Recompute recomputes coarse position phrases from the transformed training mask when a mild affine, translation, or zoom is sampled. Values are mean $\pm$ sample standard deviation. Delta and the reported bootstrap interval are MEDEQUISEG + Location Recompute minus its reference model in percentage points. Both runs use the same transformation distribution but independently initialized augmentation sequences, so the realized per-case transformations are not paired. These rows are not pooled with the main ordered configuration analysis. p_W is the exploratory, unadjusted two-sided paired Wilcoxon p -value.

Dataset	Metric	Reference model	Location Recompute	Δ pp [95% CI]	p_W
BUSI	Dice	85.74 $\pm$ 0.80	84.62 $\pm$ 0.20	-1.12 [-2.62, 0.07]	.004
	IoU	79.46 $\pm$ 0.83	78.06 $\pm$ 0.64	-1.40 [-2.70, -0.30]	.002
ClinicDB	Dice	93.18 $\pm$ 0.54	93.06 $\pm$ 0.72	-0.13 [-1.25, 1.27]	.138
	IoU	87.77 $\pm$ 0.82	87.54 $\pm$ 1.07	-0.23 [-1.94, 1.78]	.117

Presence/class-stratified prompt derangement

This post hoc sensitivity analysis reuses the 15 trained complete models without retraining. For each public dataset, the complete test-prompt multiset is deterministically permuted within target-presence and diagnostic-class strata constructed from released case identifiers and prompt fields. The permutation uses neither image pixels nor masks and preserves one-to-one prompt assignment, donor identity, stratum membership, and prompt-text consistency. It changes 3,413 of 3,426 prompt strings; the 13 unavoidable unchanged cases are BUSI normal studies whose within-stratum strings are identical. Statistics first average the three model values for each case, then use paired 10,000-repeat bootstrap intervals and Holm-adjusted Wilcoxon tests. The analysis remains a privileged-prompt diagnostic because the strata and prompts

encode released target-derived information. Exactly one deterministic donor assignment per dataset was precomputed and reused across the three checkpoints. The intervals are conditional on these assignments and checkpoints; they do not represent uncertainty across alternative valid derangements or identify a causal share for the matched attributes.

Table S13 One fixed presence/class-stratified prompt assignment. The complete prompt multiset is permuted within target-presence and diagnostic-class strata. Changed/N is the number of cases receiving a different text string. Dice values are percentages (mean $\pm$ sample standard deviation across three trained models); deltas are case-first true-minus-control effects with 95% bootstrap confidence intervals. Holm p is the adjusted paired Wilcoxon value across the five datasets. One deterministic assignment per dataset was reused across checkpoints; intervals are conditional on those assignments and do not include randomization over alternative derangements.

Dataset	Changed/N	True Dice	Stratified	Δ [95% CI]	Holm p
BUSI	65/78	85.78 ± 0.45	79.87 ± 1.05	$+5.91$ [$+2.62, +9.86$]	.038
ClinicDB	61/61	92.07 ± 0.71	88.71 ± 0.37	$+3.37$ [$+0.54, +6.96$]	.142
BUS-BRA	282/282	90.24 ± 0.14	90.16 ± 0.12	$+0.08$ [$-0.10, +0.26$]	.311
BRISC	892/892	87.18 ± 0.14	86.81 ± 0.30	$+0.37$ [$+0.13, +0.65$]	.011
COVID-19	2113/2113	84.08 ± 0.13	65.63 ± 1.10	$+18.45$ [$+17.68, +19.22$]	$< 10^{-6}$
Macro avg.	3413/3426	87.87 ± 0.17	82.24 ± 0.16	$+5.63$ [$+4.71, +6.67$]	–

Table S14 Complete per-dataset Dice and IoU comparison on five public benchmarks.

Method	Metric	BUSI	ClinicDB	BUS-BRA
<i>Image+text methods</i>				
MedCLIPSeg	Dice	86.13 ± 1.09	91.96 ± 0.07	88.47 ± 0.66
	IoU	<u>79.35 ± 1.11</u>	<u>85.64 ± 0.10</u>	80.46 ± 0.76
CausalCLIPSeg Baseline	Dice	<u>84.62 ± 1.07</u>	91.19 ± 1.10	88.87 ± 0.49
	IoU	78.13 ± 1.27	85.17 ± 1.26	<u>81.38 ± 0.64</u>
LViT	Dice	81.81 ± 0.89	91.03 ± 0.53	<u>87.66 ± 0.14</u>
	IoU	74.63 ± 1.02	84.89 ± 0.58	79.75 ± 0.15
MedEquiSeg (Ours)	Dice	<u>85.78 ± 0.45</u>	92.07 ± 0.71	90.24 ± 0.14
	IoU	79.49 ± 0.23	86.36 ± 0.61	83.15 ± 0.10
<i>Image-only contextual references</i>				
RollingUNet	Dice	72.36 ± 1.20	87.31 ± 0.77	88.00 ± 0.45
	IoU	65.06 ± 1.46	80.12 ± 0.91	80.26 ± 0.56
U-Net	Dice	61.31 ± 1.16	85.61 ± 1.28	87.35 ± 0.13
	IoU	52.70 ± 1.55	77.72 ± 1.38	79.06 ± 0.17
U-Net++	Dice	76.21 ± 0.67	89.77 ± 0.64	88.59 ± 0.06
	IoU	69.41 ± 0.79	83.20 ± 0.83	81.12 ± 0.07
nnU-Net	Dice	74.99 ± 0.23	93.62 ± 0.31	<u>89.80 ± 0.44</u>
	IoU	68.63 ± 0.27	88.63 ± 0.29	<u>82.90 ± 0.48</u>
U-Mamba	Dice	76.63 ± 0.80	92.21 ± 0.16	90.34 ± 0.46
	IoU	<u>70.16 ± 0.55</u>	<u>87.12 ± 0.33</u>	83.58 ± 0.47
U-KAN	Dice	77.10 ± 3.33	91.40 ± 0.34	88.56 ± 0.75
	IoU	70.38 ± 3.37	85.31 ± 0.33	81.19 ± 0.81
Method	Metric	BRISC	COVID-19	
<i>Image+text methods</i>				
MedCLIPSeg	Dice	81.36 ± 1.79	82.62 ± 1.19	
	IoU	72.44 ± 2.26	<u>73.47 ± 1.48</u>	
CausalCLIPSeg Baseline	Dice	86.11 ± 0.31	81.85 ± 0.15	
	IoU	<u>79.01 ± 0.51</u>	72.91 ± 0.12	
LViT	Dice	85.24 ± 0.06	78.50 ± 0.44	
	IoU	78.54 ± 0.11	69.03 ± 0.41	
MedEquiSeg (Ours)	Dice	87.18 ± 0.14	84.08 ± 0.13	
	IoU	80.32 ± 0.13	75.51 ± 0.14	
<i>Image-only contextual references</i>				
RollingUNet	Dice	85.03 ± 0.65	79.32 ± 0.24	
	IoU	77.86 ± 0.60	69.82 ± 0.34	
U-Net	Dice	85.00 ± 0.12	78.81 ± 0.07	
	IoU	77.50 ± 0.04	69.02 ± 0.12	
U-Net++	Dice	85.94 ± 0.41	80.00 ± 0.06	
	IoU	78.99 ± 0.37	70.51 ± 0.08	
nnU-Net	Dice	87.30 ± 0.35	79.07 ± 0.14	
	IoU	81.31 ± 0.43	69.67 ± 0.24	
U-Mamba	Dice	85.37 ± 0.58	79.08 ± 0.31	
	IoU	79.16 ± 0.66	69.68 ± 0.42	
U-KAN	Dice	85.45 ± 0.32	79.38 ± 0.21	
	IoU	78.33 ± 0.35	<u>69.91 ± 0.30</u>	

Notes: Values are percentages reported as mean±sample standard deviation over three seeds. Bold and underline rank methods only within the image+text or image-only panel. Cross-panel differences are descriptive because the public image+text methods receive target-derived synthetic captions unavailable to image-only methods. The five MEDEquiSEG entries use the models from the main ordered configuration analysis.

Targeted matched prompt-rewrite control

To estimate prompt rewriting under a fixed shared augmentation policy, two BUSI arms were newly trained for 100 epochs with seeds 123, 456, and 789. Within each seed pair, the fixed partition, pretrained initialization, pooled BioMedCLIP cache, model recipe, and deterministic image-mask augmentation plan were held constant; only spatial-phrase rewriting after flips and right-angle rotations differed. This targeted control is separate from the 75-run ordered matrix and does not estimate an across-dataset rewriting effect or the effect of shared augmentation itself.

Table S15 Targeted matched prompt-rewrite comparison on BUSI.

Training arm	Dice (%)	IoU (%)
Prompt rewriting disabled	84.55 ± 2.03	77.91 ± 2.25
Prompt rewriting enabled	85.71 ± 0.57	79.36 ± 0.83
Enabled – disabled	$+1.17 [-1.58, +4.17]$	$+1.45 [-1.55, +4.79]$

Notes: Both arms were newly trained for 100 epochs from the same pretrained initialization with seeds 123, 456, and 789, the same fixed BUSI partition, pooled BioMedCLIP cache, two forward-active projector ATConv operators, and identical deterministic image-mask augmentation plans. Only prompt rewriting after flips and right-angle rotations differed. Arm values are mean $\pm$ sample standard deviation across seeds. Brackets give the 95% hierarchical bootstrap interval for the percentage-point difference after resampling both test cases and training seeds. This targeted BUSI comparison does not estimate an across-dataset or clinical effect.

Complete-minus-Base statistical sensitivity

This secondary analysis reuses the existing complete-configuration and Base predictions. It resamples paired test cases and the three paired training-seed indices hierarchically and repeats the equal-dataset macro-average after excluding empty reference masks. It does not retrain any model or add independent training seeds.

Table S16 Secondary complete-minus-Base sensitivity analyses

Analysis set	Dice, pp (95% CI)	IoU, pp (95% CI)
All test cases	+1.46 [+0.76, +2.29]	+1.74 [+1.02, +2.55]
Non-empty reference masks	+1.53 [+0.80, +2.37]	+1.84 [+1.06, +2.68]

Differences are complete configuration minus Base and use an equal-dataset macro-average. Intervals are secondary hierarchical paired bootstraps (10,000 replicates) that resample test cases within dataset and the three training-seed pairs. Empty status is determined from the dataset-provided reference mask. These analyses reuse existing predictions and do not add independent training seeds or retrain any model.